

A separably represented C*-algebra with no separable irreducible representation

MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation

theorem

The same algebra T has a faithful representation on a separable Hilbert space, but no nonzero irreducible representation on any separable Hilbert space.

Statement

Let A be the completed CAR algebra. Let $\pi : A \rightarrow B(\mathcal{H})$ be the direct sum of one chosen GNS representation from each pure-state GNS equivalence class, and let $(L_i)_{i \in I}$ be the unitaries obtained from the fixed CAR homogeneity and shell construction. Put $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$, and write $\iota(a) = \pi(a)$ for the embedding into T . Let τ be the normalized trace of A . The algebra T retains all the properties in the cited representation-classification theorem: it is nonzero, norm closed, unital, infinite dimensional and simple; ι is injective and unital; the inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ is faithful and irreducible; every nonzero irreducible $*$ -representation of T is unitarily equivalent to ρ_0 ; and T is not isomorphic to the full algebra of compact operators on any Hilbert space. In addition, $H_\tau = L^2(A, \tau)$ is nonzero and separable, and there exists a faithful unital representation $T \rightarrow B(H_\tau)$. No separable complex Hilbert space carries a nonzero irreducible $*$ -representation of T , even without assuming preservation of the identity.

Assumptions

All assertions concern the same generated algebra T and the same fixed shell unitaries. The absence of irreducible representations is quantified over arbitrary separable complex Hilbert spaces. Norm separability of T is neither required nor asserted.

Conclusion

T has both the faithful representation $\tilde{\rho}$ on the separable space H_τ and the faithful irreducible representation ρ_0 on $\mathcal{H}$. The representation $\tilde{\rho}$ is reducible, whereas $\mathcal{H}$ cannot be separable.

The faithful representation in the existential assertion is the already chosen $\tilde{\rho}$. No different algebra or unrelated representation is chosen to establish the two separability assertions.

Proof route

Combine the classification and simplicity results for T with the normalized CAR trace vector, the dense CAR orbit, faithfulness of $\tilde{\rho}$, and the theorem excluding nonzero irreducible representations on separable Hilbert spaces.

▼ Proof steps

1. Let $(\pi_\tau, H_\tau, \xi_\tau)$ be the GNS representation and cyclic vector of τ . One has $\|\xi_\tau\| = 1$, so $H_\tau \neq \{0\}$. The continuous map $a \mapsto \pi_\tau(a)\xi_\tau$ has dense range. Since A is separable, H_τ is separable.

2. Take the GNS representation ρ_ϕ of the chosen extension ϕ of τ to T , and the unitary $E : H_\tau \rightarrow H_\phi$ from the cited trace construction. Then $\tilde{\rho}(t) = E^{-1} \rho_\phi(t) E$ is the established faithful representation on H_τ . The cited theorem excluding nonzero irreducible representations on separable Hilbert spaces applies to every $*$ -representation $\rho : T \rightarrow B(H)$ with H separable. These statements are conjoined with the classification and structural properties of the same T .

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation
- The classification and structural properties of the constructed algebra —
MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint
- The fixed trace-space abbreviation —
MathlibAnnex.CStarAlgebra.CAR.SeparableCounterexampleHilbertSpace
- A nonzero vector in the trace space —
MathlibAnnex.CStarAlgebra.CAR.nontrivial_traceHilbertSpace
- Separability of that same space —
MathlibAnnex.CStarAlgebra.CAR.separableSpace_separableCounterexampleHilbertSpace
- The faithful witness —
MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective
- Exclusion on arbitrary separable Hilbert spaces —
MathlibAnnex.CStarAlgebra.CAR.not_isIrreducible_of_separable

Supporting route explanation

Lean source signature (exact)

```
theorem
atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation :
  AtomicCounterexampleEndpoint.{v} ∧
  Nontrivial SeparableCounterexampleHilbertSpace ∧
  TopologicalSpace.SeparableSpace SeparableCounterexampleHilbertSpace ∧
  (∃ ρ : Representation AtomicCounterexampleAlgebra
SeparableCounterexampleHilbertSpace,
    Function.Injective ρ) ∧
  (∀ (H : Type v) [NormedAddCommGroup H] [InnerProductSpace ℂ H]
    [CompleteSpace H] [TopologicalSpace.SeparableSpace H],
    ∀ ρ : NonUnitalRepresentation (A := AtomicCounterexampleAlgebra) (H := H),
      ¬ ρ.IsIrreducible)
```

In the source

AtomicCounterexampleEndpoint.{v}

Mathematical meaning

First output: all ten structural properties of the same fixed target T , source ι , and inclusion ρ_0 . The separate

In the source	Mathematical meaning
	exact abbreviation, structure and field table below make this component explicit within this Card.
Nontrivial SeparableCounterexampleHilbertSpace	Second output: the CAR trace GNS space $H_\tau = L^2(A, \tau)$ is nonzero.
TopologicalSpace.SeparableSpace SeparableCounterexampleHilbertSpace	Third output: this same H_τ has a countable norm-dense subset.
$\exists p : \text{Representation}$ AtomicCounterexampleAlgebra SeparableCounterexampleHilbertSpace, Function.Injective p	Fourth output: there exists an injective unital star representation $T \rightarrow B(H_\tau)$, supplied by the fixed $\hat{\rho}$. It is faithful but is not asserted irreducible.
$\forall (H : \text{Type } v) [\text{NormedAddCommGroup } H]$ [InnerProductSpace $\mathbb{C} H]$ [CompleteSpace $H]$ [TopologicalSpace.SeparableSpace $H]$	For the fifth output, now quantify any separable complete complex Hilbert space H in universe v ; it is not restricted to the previously fixed H_τ .
$\forall p : \text{NonUnitalRepresentation } (A := \text{AtomicCounterexampleAlgebra}) (H := H), \neg$ $p.\text{IsIrreducible}$	For every possibly nonunital star representation on that H , nonzero irreducibility fails. This universal ρ is different in quantifier role from the fourth output's existential representation. None of these outputs asserts norm separability of T .

These are two separate exact related declarations: the result abbreviation (AtomicCounterexample.lean, lines 92–93), then its ten-field proposition (AtomicEndpoint.lean, lines 194–229). Neither is a replacement or extension of the theorem signature above. Here the family is the fixed homogeneityShellFamily, so T, ι, ρ_0 have exactly the meaning given in this Card.

```
abbrev AtomicCounterexampleEndpoint : Prop :=
  ShellFamilyEndpoint.{v} homogeneityShellFamily
```

```
structure ShellFamilyEndpoint (family : RepresentativeShellFamily) : Prop where
  nontrivial_target : Nontrivial (ShellFamilyTarget family)
  isClosed_target :
    IsClosed
      (ShellFamilyTarget family : Set
        (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
          → L[ $\mathbb{C}$ ]
            MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
              completedRootPureState))
  source_injective : Function.Injective (shellFamilySourceHom family)
  source_unital : shellFamilySourceHom family 1 = 1
```

```

not_finiteDimensional_target :
  ¬ FiniteDimensional ℂ (ShellFamilyTarget family)
ambient_injective : Function.Injective (shellFamilyInclusion family)
isIrreducible_ambient :
  Representation.IsIrreducible (shellFamilyInclusion family)
captures_nonunital :
  ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace ℂ K]
    [CompleteSpace K]
    (rho : NonUnitalRepresentation
      (A := ShellFamilyTarget family) (H := K)),
    rho.IsIrreducible →
      ∃ U :
        MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState ≈li[ℂ] K,
  ∀ (a : ShellFamilyTarget family)
    (x : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState),
    U (shellFamilyInclusion family a x) = rho a (U x)
closedIdeal_dichotomy :
  ∀ I : TwoSidedIdeal (ShellFamilyTarget family),
    IsClosed (I : Set (ShellFamilyTarget family)) → I = ⊥ ∨ I = T
not_compactOperatorModel :
  ∀ (K : Type v) [NormedAddCommGroup K] [InnerProductSpace ℂ K]
    [CompleteSpace K]
    (e : ShellFamilyTarget family →na[ℂ] (K →L[ℂ] K)),
    ¬ (Function.Injective e ∧
      (∀ a : ShellFamilyTarget family, IsCompactOperator (e a)) ∧
      ∀ T : K →L[ℂ] K, IsCompactOperator T →
        ∃ a : ShellFamilyTarget family, e a = T)

```

In the source	Mathematical meaning
nontrivial_target	The fixed algebra T is nonzero.
isClosed_target	The set $T \subseteq B(\mathcal{H})$ is closed for the operator norm.
source_injective	The fixed source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, is injective.
source_unital	The same map sends 1_A to 1_T .
not_finiteDimensional_target	T is not finite-dimensional over $\mathbb{C}$.

In the source	Mathematical meaning
<code>ambient_injective</code>	The literal inclusion $\rho_0 : T \rightarrow B(\mathcal{H})$ is injective.
<code>isIrreducible_ambient</code>	This same ρ_0 is nonzero irreducible: it has no proper nonzero closed reducing subspace.
<code>captures_nonunital</code>	After the family and T have been fixed, for every complete complex Hilbert space K in universe v and every nonzero irreducible possibly nonunital $\rho : T \rightarrow B(K)$, there is a surjective complex-linear isometry $U : \mathcal{H} \rightarrow K$ with $U(\rho_0(t)x) = \rho(t)Ux$ for every $t \in T, x \in \mathcal{H}$.
<code>closedIdeal_dichotomy</code>	For every two-sided ideal $\mathfrak{a}$ in T , if its underlying set is norm closed, then $\mathfrak{a} = \{0\}$ or $\mathfrak{a} = T$.
<code>not_compactOperatorModel</code>	For every complete complex Hilbert space K and every possibly nonunital star homomorphism $e : T \rightarrow B(K)$, it is impossible that e is injective, all $e(t)$ are compact, and every compact operator on K equals some $e(t)$. The zero space is included.

Further source notes: The long conjunction in the exact signature keeps `AtomicCounterexampleEndpoint`, `Nontrivial` and `SeparableSpace` of H_τ , an existential injective representation, and the universal exclusion statement separate. `SeparableCounterexampleHilbertSpace` is H_τ and `NonUnitalRepresentation` does not initially require $\rho(1) = 1$. The proof chooses `separableCounterexampleRepresentation`, the same $\tilde{\rho}$.

Earlier published Card and PDF

▼ Exact content identity

Declaration:
 MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation
 Accepted content SHA-256: df641d48577ed178651ff18b5cdc2c3fb4e0d88c8cf4fd6d01b96628ed25d648
 Accepted source guide SHA-256: 16f450a005ba142d889b4ec3737f5ca01b0d44a0d907db676530e2f74356bbf2
 Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73